

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15 06
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks:
40

Duration :60 Minutes

Code of Conduct

I **EDIGA KHADRI NATH GOWD / 192472296** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **EDIGA KHADRI NATH GOWD**

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Industry Problem Solving Task

1. Retail Data Optimization Problem

Proposed Big Data Architecture

The retail chain can use a Lambda/Kappa-style architecture to process both real-time and historical data.

Data Sources:

- Point-of-Sale (POS) systems
- E-commerce transactions
- Inventory systems
- Customer browsing and purchase history
- Mobile applications

Data Ingestion:

- Apache Kafka for real-time sales, inventory, and customer activity streams.
- Apache NiFi or ETL tools for batch data collection.

Storage:

- HDFS / Amazon S3 / Azure Data Lake for storing large historical datasets.
- MongoDB or similar NoSQL databases for flexible customer and product data.

Processing:

- Apache Spark for large-scale batch and streaming data processing.
- Spark Structured Streaming for real-time inventory updates.

Analytics and ML Models:

- Time-series forecasting: ARIMA, Prophet, or LSTM for demand forecasting.
- Recommendation: Collaborative Filtering and Association Rule Mining.
- Clustering: K-Means for customer segmentation.

Output:

Dashboards can be developed using Power BI/Tableau to show inventory levels, predicted demand, sales trends, and customer recommendations.

2. Social Media Fraud Detection Problem

Proposed Big Data Solution

Data Sources:

- Posts
- Comments
- Likes
- Shares
- User profiles
- Login and activity information

Data Ingestion:

- Apache Kafka collects billions of user interactions in real time.

Stream Processing:

- Apache Spark Structured Streaming or Apache Flink processes incoming events with low latency.

Storage:

- HDFS/S3 stores historical social media data.
- NoSQL databases such as Cassandra can store high-volume user activity.

Machine Learning:

- Random Forest/XGBoost for fake-account classification.
- Isolation Forest for detecting unusual user behavior.
- Natural Language Processing (NLP) models such as BERT can analyze posts for misinformation.
- Graph-based analysis can identify suspicious connections between accounts.

Process:

Social Media → Kafka → Spark/Flink → ML Models → Fraud/
Misinformation Alert → Dashboard

This architecture provides scalable and near real-time fraud detection.

3. Government Data Platform Problem

Proposed Governance and Data Management Strategy

The government should create a centralized national data platform using a secure data lakehouse architecture.

Data Sources:

- Census databases
- Economic datasets
- Geospatial/GIS data
- Department databases

Data Integration:

- Apache Kafka for real-time data exchange.
- ETL/ELT pipelines for batch integration.
- APIs should be used for interoperability between departments.

Storage:

- Cloud Data Lake such as Amazon S3/Azure Data Lake.
- A data warehouse/lakehouse can provide structured analytical access.

Governance:

- Establish common data standards and formats.
- Maintain a data catalog and metadata management system.
- Define data ownership and access policies.
- Implement data quality and validation rules.

Security and Privacy:

- Role-Based Access Control (RBAC)
- Encryption during storage and transmission
- Multi-factor authentication
- Data masking and anonymization
- Audit logs for monitoring access

This approach enables secure data sharing while maintaining interoperability, privacy, and accountability.

4. Banking Fraud Detection Problem

Proposed Big Data Pipeline

A bank needs a real-time streaming architecture capable of processing millions of transactions with very low latency.

Data Sources:

- ATM transactions
- Credit/debit card transactions
- Online banking
- Mobile banking
- Historical customer transactions

Pipeline:

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Score → Alert/Block

Technologies:

- Apache Kafka: Real-time transaction ingestion.
- Apache Flink/Spark Streaming: Real-time processing.
- HDFS/S3: Historical transaction storage.
- Spark: Large-scale historical analysis.
- Redis/Cassandra: Fast access to customer and transaction features.

Algorithms:

- Random Forest/XGBoost: Fraud classification.
- Logistic Regression: Baseline fraud prediction.
- Isolation Forest: Anomaly detection.
- Neural Networks: Complex fraud pattern detection.

The system can calculate a fraud score for each transaction and immediately generate an alert or block suspicious transactions.

5. Agriculture Smart Farming Problem

Proposed Big Data Solution

A precision agriculture architecture can combine IoT sensors, drones, weather systems, and satellite data.

Data Sources:

- Soil moisture sensors
- Temperature and humidity sensors
- Weather stations
- Drones
- Satellite images
- Crop and historical yield data

Data Ingestion:

- IoT gateways collect sensor data.
- MQTT/Kafka can transmit real-time sensor information.

Storage:

- Cloud Data Lake stores sensor, image, weather, and historical data.
- NoSQL databases can store high-volume sensor data.

Processing:

- Apache Spark processes large datasets.
- Stream processing provides real-time monitoring.

Analytics and ML:

- Regression models: Predict crop yield.
- Random Forest: Crop disease and risk prediction.
- CNN: Analyze drone/satellite images for disease detection.
- Time-series models: Predict weather and irrigation requirements.
- Clustering: Identify different soil/crop zones.

Decision Support:

The system can provide recommendations for:

- Irrigation

- Fertilizer usage
- Pest control
- Crop selection
- Harvest timing

This reduces resource wastage, improves crop yield, and supports data-driven farming decisions.